

FOLLOWUP_ITEMS

BOB-207 — follow-up items, drafted and ready to file

Revision: 1 Last modified: 2026-08-27T00:00:00Z

Why these are drafted rather than filed

§11.4.202 requires a reported item to land in the SQLite SSoT **and** drive the derived documents back into sync (docs/Issues.md, docs/Issues_Summary.md). docs/workable_items.db, docs/Issues.md and docs/Issues_Summary.md all carry uncommitted modifications from prior work, three work streams are live in this checkout, and this stream is under a no-commit/no-stage constraint — so it cannot complete the DB→docs regeneration cycle. Filing the DB row alone would leave the tracker internally divergent, which §11.4.186 gates against.

Highest allocated id at drafting time: **BOB-225**. Ids below are placeholders to be assigned at filing.

ITEM 1 — repair-side walk over foreign-owned INTERIOR directories is untested

Type: Task · **Status:** Queued · **Severity:** medium **Layer:** integration
(the unprivileged-harness constraint does not bind there)

WHAT. tests/unit/test_ownership_repair.sh seeds the foreign uid onto **files and symlinks only**; interior directories stay operator-owned. Case 22 seeds a foreign declared **root** (seed_wrong -D), but only

on the *failure* path — a shimmed chown that must fail — so no case walks a tree whose **interior** directories carry a foreign uid and succeeds.

WHY IT MATTERS. Production first-start repair faces exactly that shape: research.md R6 measured the download root and config/ with foreign-owned directories throughout. The repair must chown those directories, and on a directory the kernel does **not** clear S_ISGID on chown(2) (see ITEM 2), so directory repair has behaviour that file repair does not.

NOT ALREADY COVERED. tests/ownership/test_container_writes_owned_files.py::test_container_written_directory_is_owned_by_the_operator covers the **creation** side (FR-002) — a container writing a directory. It does not exercise the **repair** walk over a pre-existing foreign-owned directory tree. The DECLARED GAPS bullet in the unit suite has been tightened so it can no longer be read as claiming repair-side coverage.

WHY UNIT-LEVEL IS THE WRONG HOME. Seeding interior directories foreign would leave the harness unable to create the symlinks Cases 8/9/18 add *after* seeding — an operator cannot write into a directory they no longer own. The integration layer has no such constraint.

ACCEPTANCE. An integration test seeds a tree with foreign-owned interior directories, runs scripts/ownership_repair.sh to completion, and asserts every directory ends operator-owned, with a paired \$1.1 mutation that makes it fail.

FR refs: FR-002, FR-004, FR-012 · **Discovery channel (§11.4.238):** independent code review of the BOB-207 re-base, not the automated regime.

ITEM 2 — directory setgid is silently dropped by the mode-restore mask

Type: Bug · **Status:** Queued · **Severity:** medium

WHAT. scripts/ownership_repair.sh:892-893 masks the restored mode with 8#1777. Its in-source rationale is that the kernel clears S_ISUID/S_ISGID on chown(2), so the high bits cannot be restored anyway. That holds for **files**. It does **not** hold for **directories**: Linux does not clear

S_ISGID on a directory chown(2), and on a directory S_ISGID means group inheritance for newly created children. A foreign-owned 2755 directory repaired through that path therefore loses its inheritance bit, silently.

PRE-EXISTING, NOT INTRODUCED. Byte-identical at HEAD a3e1141777914f3d71d62a6cd32a4c79c7f43842; outside the BOB-207 change.

BOB-207 SHRINKS THE HAZARD. With the walk narrowed to uid, a setgid directory the operator **already owns** is never selected, so it is never chowned and never mode-restored. The exposure is now limited to directories that are *both* foreign-owned *and* setgid — the ITEM 1 shape.

ACCEPTANCE. Decide (operator, §11.4.66) whether directory setgid must survive repair. If yes: preserve S_ISGID for directories, with a test that fails against the current mask. If no: record the decision in-source so the next reader does not re-derive the question.

FR refs: FR-015, FR-005 · **Related:** ITEM 1 (same directory shape).